

The Syntax of Orientation Shifting in English High Adverbs

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1 Introduction

It has previously been noted that some ‘high’ adverbs in the sense of Cinque (1999) shift orientation in questions compared with declarative sentences. The orientation of an adverb refers to the individual whose attitudes, opinions or knowledge the adverb expresses. The individuals in question here are the discourse participants, that is the speaker and the addressee involved in the speech act. Illocutionary and evidential adverbs orient to the speaker of the sentence in declarative sentences but in questions orient to the addressee. However, the syntax and semantics of this phenomenon have not yet been fully analysed despite its consistency and systematicity.

This paper has two aims¹: first, to show empirically the contexts in which adverbs shift orientation. Second, to analyse how this shift occurs using independently motivated principles. It will be proposed that discourse participants are syntactically represented in a covert speech act verb phrase above matrix CP, and that the high adverb has a logophorically-sensitive PRO subject which is controlled by the appropriate discourse participant. This analysis will not only be shown to be elegant but also in the spirit of the Minimalist Program (Chomsky 1995).

The paper is structured as follows: section 2 outlines previous work on adverb orientation and orientation shifting and section 3 presents the key data. Section 4 presents a syntactic analysis of the phenomenon, with further detail on the syntax and semantics of speech acts in section 5. Section 6 will bring together my conclusions with directions for further research.

2 Previous research

Two areas of research will be outlined in this section: firstly, work concerning the orientation of adverbs. This will include a brief overview of work on grammaticalised elements such as evidentiality markers which also shift in interrogatives. Secondly, key research on the representation of discourse participants in syntax will provide a context for the analysis to follow.

¹This paper was first submitted for the confirmation of candidature procedure at the Department of Language and Linguistic Science, University of York in September 2013. A condensed version was also presented at ConSOLE XXII, University of Lisbon, in January 2014.

Jackendoff (1972) first argued that adverbs are a part of speech in their own right rather than the product of a transformation rule applied to nominals. He identified three types of adverb based primarily on the surface syntactic position of these adverbs in sentences, but also based on their orientation. Jackendoff defines speaker orientation as follows (1972, p.56):

Some [adverbs] are understood as relating the speaker's attitude toward the event expressed by the sentence.

He demonstrates the orientation of such adverbs using a variety of paraphrases (examples from Jackendoff 1972, pp.56-58):

- (1) a. Frankly, John lied to Bill
- b. *John lied to Bill frankly
- c. Frankly speaking, John lied to Bill
- d. To tell the truth, John lied to Bill

He concludes that speaker-oriented adverbs predicate over sentences and speakers from sentence-initial position, whereas subject-oriented and VP adverbs are more deeply embedded within the sentence itself.

Building upon his work, Bellert (1977) notes that Jackendoff's 'speaker-oriented' adverbs do not constitute a homogeneous group but vary in their purpose and structure. In particular, she shows that syntactic position alone does not adequately characterise different adverb classes; instead, she divides adverbs into classes which foreshadow Cinque's (1999) categories according to the number and type of semantic arguments they take. Of particular importance are those adverbs which she defines as taking two arguments, namely pragmatic, evaluative, and modal adverbs.

Evaluative adverbs such as *luckily*, *fortunately*, *happily* and so on take the event or state of affairs denoted by the sentence as an argument, as well as an evaluation of this event or state of affairs. These propositions are asserted and can be negated independently:

- (2) (Bellert's (13) and (14), p.342)
- a. Fortunately, John has come $\rightarrow$ John has come
- b. Unfortunately, John has come $\rightarrow$ John has come
- c. Unfortunately, John has not come $\rightarrow$ John has not come

Bellert notes that their status as assertions is the reason why evaluative adverbs are not available in questions, because it is not possible to question one proposition while simultaneously asserting a second related proposition (example from Bellert 1977, p.343):

- (3) (Bellert's (15), p.343)
- Has John unfortunately arrived?

Modal adverbs (corresponding to Cinque’s evidential and epistemic adverbs) in Bellert’s account do not take the proposition of the sentence but the truth of the proposition of the sentence as one argument. Examples include *probably*, *certainly*, *evidently*, and so on. Such adverbs do not have negative counterparts and cannot be negated independently of the sentence:

- (4) (from Bellert’s (17), p.343)
 - a. *Improbably, John has come (meaning: It is improbable that John has come)
 - b. *Not evidently, John has come.

They are also above sentential negation:

- (5) (from Bellert’s (30) and (32), p.346)
 - a. John probably never ran so fast.
 - b. *Never did John probably run so fast.

They can also occur in if-then clauses, which is not true of evaluative adverbs. Bellert notes as a further consequence that modal adverbs do not occur in questions because an evaluation of the truth cannot be made of a proposition while that proposition is being questioned. However, she concedes that some modal adverbs with ‘an additional meaning component’, such as *definitely* and *perhaps* can occur in questions as they imply a particular answer:

- (6) (Bellert’s (22) and (24), p.344)
 - a. Has John perhaps been here before?
 - b. Has John definitely made up his mind?

Finally, Bellert notes that only pragmatic adverbs - Cinque’s illocutionary adverbs - are truly speaker-oriented in that they take the speaker as an argument. These adverbs include *frankly*, *sincerely*, *honestly* and so on. The other argument is the proposition, or content, of the sentence. As Bellert explains, “The speaker characterizes his attitude towards *what* he is saying” (p.349, original emphasis). Correspondingly, these adverbs also do not have negative counterparts (that may be used as sentential adverbs) and are also above sentential negation. Furthermore, these adverbs are possible in non-asserted sentences, such as performative sentences and questions.

Bellert refines Jackendoff’s criteria carefully using convincing semantic arguments. Her work is also important as an early example of analyses which consider adverbs to contain some kind of argument structure; she even entertains the idea of a kind of performative analysis in the style of Ross (Bellert 1977, p.349f). This is an important precedent for the account put forward in this paper, though it will be shown that the modal adverb category can be divided further and the behaviour of adverbs which occur in questions requires further consideration.

Coming from a literary perspective, Banfield (1982) examines adverb orientation in the context of Free Indirect Discourse (FID). In FID the speaker (read: writer) and the mind expressed in the discourse (the literary character from whose point of view (PoV) the text is written) are dissociated. She shows that the literary character's PoV supersedes the PoV of the writer because indexical, temporal and deictic elements relate to the literary character instead of the writer. The fact that the writer and the third-person character's PoVs are mutually exclusive is key for Banfield's analysis of FID. However, there seem to be some elements which are irrevocably linked to the 'true' or 'real' discourse authors. For example in (7a), the illocutionary adverb *honestly* is not readily interpretable with a third-person PoV despite the literary character superseding the writer. In contrast, the epistemic adverb in (7b) is interpretable with a third-person (literary character) PoV (examples from Banfield 1982, p.117):

- (7) a. Honestly, she was so pleased to see him - delighted!
- b. Certainly, she was so pleased to see him - delighted!

This contrast between first- and third-person selves is related to Banfield's conceptualisation of direct and indirect quotation. To use direct quotation, she says, "is to 'donner la parole' to the original speaker", and is expressed in the first person. Indirect speech, however, represents only the content of the speech and not the way in which it has been delivered (Banfield 1982, p.62). Like FID, indirect speech is expressed in the third person. However, it will become clear that direct speech-like environments can still be induced in indirect speech contexts, which will show that the divorce between 'speaker' and 'self' applies to more contexts than just FID (cf. also Broadwell (1991)). This shall be detailed in section 4.2.2.

The question of how different clause types cause shifts in discourse participants' PoVs has been examined by scholars such as Garrett (2001) in relation to grammaticalised evidentiality markers. Garrett analyses evidential particles in Tibetan, noting that they are speaker-oriented in declarative sentences but addressee-oriented in questions; in his words, "the origo [ego] of a question is always the person to whom the question is asked." (2001, p.230). However, he also asserts that the subject of an embedded question must be the perspective holder of an embedded question. Both statements are true in Garrett's data as the subjects of embedded questions are always coreferential with the matrix object (the addressee of the matrix verb of communication). Garrett's data set therefore underdetermines the phenomenon, as the adverb in fact never orients to the embedded subject except in a very narrow set of cases. However, the analysis below will show that there are structural parallels between adverb orientation in matrix and embedded clauses.

Garrett suggests that the origo shift occurs because the hearer is 'inside' the question and opts for a partially semantic, partially pragmatic approach to explain this. Building on Hamblin (1973) and Karttunen's (1977)'s analyses of questions as sets of possible or true answers, Garrett proposes that the answer-set consists not of propositions, but of asserted speech acts made from

the hearer’s perspective. Moreover, he represents the hearer and speaker as free variables which form an ordered set of participants in the assertive speech act. Therefore, where Hamblin would give the denotation of “Can you play football?” as a set of {You can play football, You cannot play football}, Garrett proposes something like {“I tell you that I can play football”, “I tell you that I cannot play football”}.

However, this is the extent of Garrett’s semantic analysis. He suggests that these variables and their relationship with the evidential markers are determined pragmatically through context alone, and that “the origo shift comes for free as long as assertions have authors and questions have answerers” (2001, p.250). This reliance on pragmatics seems at odds with the fact that the origo shift is obligatory in Tibetan (cf. De Villiers et al 2009) and yet it is not clear exactly how he conceptualises the discourse participants and their relationship with evidentiality in his semantic account.

Clearly, a successful analysis of shifting must also explain how the discourse participants are represented in the sentence and the relationships between them. Several syntactic and semantic analyses of the representation of discourse participants have so far been proposed, of which some are detailed below.

It was first noted by philosophers that there are parallels between typical declarative sentences and clauses embedded under performative verbs; Austin (1962, p.32) proposed that every illocutionary act contains a performative verb in some form. Following Austin’s work, Ross (1970) suggested that speakers and addressees were covertly represented in syntax, at least in the context of performative verb constructions. He proposed this to account for a range of phenomena, most notably cases of reflexive pronouns which are grammatical despite lacking an overt antecedent, as in the following examples (from Ross 1970):

- (8) a. This paper was written by Ann and myself (p.228)
- b. As for myself, I won’t be invited (p.232)

Ross also presents evidence from deleted arguments. He claims that in some dialects the following examples are acceptable and that the dropped arguments implicitly relate to a first-person discourse participant:

- (9) a. Sid is coming with² ~~me/us/*you/*him~~ (p.236)
- b. A friend of ~~mine/of ours/*of hers~~ is going to drop by (p.238)

Ross presents two possible solutions to this problem. His first solution, based on the behaviour of performative verbs, is that there is a phonologically null performative verb above all declarative sentences. The subject of this verb is first-person and the object second-person (Ross 1970, p.252). This is a purely syntactic solution which does not rely on semantic or pragmatic principles and

²The accent here represents prosodic emphasis which often occurs in this construction.

which only extends to affirmative declarative clauses.

His second solution, which he terms the ‘pragmatic analysis’, rests on an early interpretation of the syntax-discourse interface. Ross suggests that elements “in the air”, i.e. in the discourse context, are available antecedents just like any element in the deep structure (Ross’s terms) and, by extension, relations between the elements “in the air” and those in the syntax are constrained by the same kind of hierarchical rules which constrain syntactic relations. Ross suggests that the pragmatic analysis might be preferable to the syntactic analysis as the former does not have any of the ‘tension’ of the latter in terms of matrix clauses being technically embedded beneath the covert performative phrase.

However, Ross asserts that any elements called upon from the context would have to be “hierarchically grouped to form a structure which is exactly the same as that of a normal clause in deep structure” (1970, p.255). Although it would be possible to take elements from context as antecedents, it might not be possible to order them hierarchically as would be necessary to ensure the correct antecedent for a syntactic element. Therefore, an account of this kind could not be purely pragmatic but would have to call upon syntactic principles to form the necessary hierarchical structure.

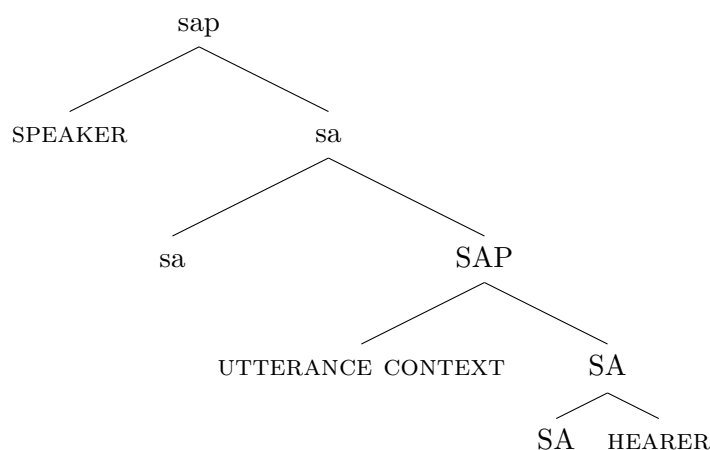
Ross’s analyses were largely abandoned during the Government & Binding period in generative linguistics. However, with the advent of the Minimalist Program (Chomsky 1995), particularly phase-based iterations thereof, other scholars have proposed similar analyses. Tsoulas and Kural’s (1999, hereafter T&K) work, motivated by the observation that the reference of indexicals shifts according to the utterance context, proposes that indexical pronouns are bound variables. Not all indexicals shift; T&K identify ‘I’, ‘you’ and temporal expressions such as ‘now’ or ‘today’ as examples of indexicals which shift, in contrast with third-person pronouns and deictic expressions such as ‘that day’, whose reference does not change. They propose that the reference of shifting indexicals is established through an operator-variable relationship between (for example) covert SPEAKER and ADDRESSEE operators and first- and second-person pronouns as variables. They show that high adverbs take scope over variables to give a generic reading, which shows that indexicals do not refer directly in the same way that names do. Furthermore, the presence of covert SPEAKERS and ADDRESSEES can account for PoV phenomena on relational nouns; for example in the sentence “Sue saw Mom at the airport”, ‘Mom’ can only refer to the mother of the speaker (and possibly addressee, if speaker and addressee are siblings), but it cannot refer to Sue’s mother (T&K 1999, p.555). It must therefore be bound by a representation of the speaker (and addressee), such as the operators proposed by T&K.

T&K note that these operators will be syntactically represented above matrix CP, though they are not explicit about where or how they fit into the structure. They also do not state whether the operators are always present or whether they only occur in certain types of sentences.

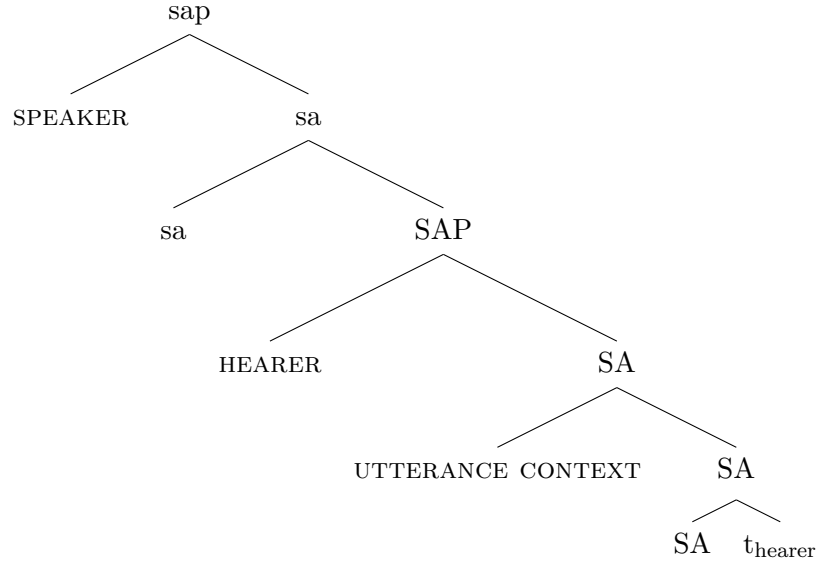
Both questions are addressed by Speas & Tenny (2003, hereafter S&T), who envisage a syntactic representation of T&K's ideas which brings together the intuition of discourse participants in the syntax with established restrictions on the role of pragmatic forces within grammar. S&T suggest, as did Ross, that syntactically-represented discourse participants are hierarchically organised. However, they refute the idea that certain speech acts are tied to certain clause types, as questions can be answered with questions and declarative sentences can encode implicit requests and so on. This casts doubt on whether Ross's analysis should be restricted to performative declarative sentences after all, as there are many different types of sentence which are subject to illocutionary force and which therefore have implications for speakers and hearers. As support for this view, S&T observe that Cinque's (1999) SpeechAct projection, which encodes important aspects of discourse participants' involvement in the speech act, is overt in any language which contains elements such as interrogative morphemes and sentence particles, which of course are not limited to (or in some cases, ever permitted in) performative declarative clauses (S&T 2003, p.317).

S&T propose that the speaker and the addressee are represented as arguments within the extended SpeechActPhrase along with an operator-like representation of the utterance context (UC), and that this SpeechAct domain is above the sentence itself. The motivation for this is that these discourse roles are grammaticalised in some languages, such as in Basque allocutive agreement. They represent a declarative structure as in (10) and an interrogative one in (11).

(10)



(11)



Note that the difference between the two structures is that the hearer moves up to adjoin to SAP in interrogatives.³ S&T motivate the declarative configuration on the idea of the UC as the theme and the hearer as the goal of the utterance, though this would typically see the UC generated as the complement to SA (cf. Bruening (2010)). Furthermore, it is not clear how this Speech Act phrase relates to the (rest of the) matrix clause as long as ternary branching is not permitted. This is particularly important given that the logophoric role which S&T postulate, the ‘Seat of Knowledge’, is represented in [Spec, EvaluativeP], which is considered to be in the matrix clause. These criticisms do not in any way undermine the theoretical advances made by S&T on the previous literature, but this paper shall address these questions in later sections.

3 Adverb orientation: the phenomenon

3.1 Key terms

Turning now to the empirical data, adverbs which shift orientation constitute two of the four ‘high’ adverb categories identified by Cinque (1999), namely illocutionary and evidential adverbs. These terms will be used throughout. I will adopt Jackendoff’s (1972) definition of adverb orientation insofar as the adverb describes the attitude(s) *or* state of knowledge of an individual. However, these adverbs do not always describe the attitude(s) of the *speaker*. Focusing on yes-no interrogatives uttered as straightforward interrogative acts,⁴ the data show that adverbs in this kind of interrogative clause describe the attitude/state of knowledge of the addressee as ascribed to him/her by the

³sap and SAP are both SpeechAct projections, with sap as a shell to SAP.

⁴Groenendijk & Stokhof (1994, p.1055) make the distinction between syntactic interrogative clauses, pragmatic interrogative acts, and semantic questions. The focus will be on syntactically interrogative non-rhetorical yes-no questions involving inversion.

speaker, or the attitude with which the speaker expects the addressee to respond. These adverbs are therefore addressee-oriented in such contexts. Furthermore, these adverbs in clauses embedded under a verb of communication orient to the matrix subject in the case of embedded declaratives or the matrix object in the case of embedded interrogatives. This is because the individual whose situation contains the epistemically relevant information is different in declarative and interrogative clauses (see section 5).

3.2 Data

The orientation of illocutionary and evidential adverbs shifts consistently and systematically from SPEAKER in declaratives to ADDRESSEE in yes-no interrogatives:

(12) Illocutionary adverbs

- a. Seriously_{speaker}, Andy can play rugby.
- b. Seriously_{addressee}, can Andy play rugby?

(13) Evidential adverbs

- a. Allegedly_{speaker}, Ahmad is at the top of the list.
- b. Is Ahmad allegedly_{addressee}⁵ at the top of the list?

(14) Epistemic adverbs

- a. Ewan has probably_{speaker} gone to the dentist.
- b. ?*Has Ewan probably gone to the dentist?
- c. Ella is definitely_{speaker} at the top of the list.
- d. Is Ella definitely_{addressee} at the top of the list?

As (14b) shows, epistemic adverbs are ungrammatical in questions in English. They are not available for shifting and so may only take a speaker-oriented reading. Furthermore, as noted by Bellert, the epistemic adverbs *definitely* and *perhaps* are different from other epistemic adverbs as they are available in questions and also shift. These facts even hold when the context would seem to favour the opposite interpretation:

(15) Illocutionary adverbs

Context: The manager of a rugby team is likely to lose his job soon as his team has been performing badly. This weekend's match will determine his future but he is a player short and asks a friend (who doesn't care about rugby) for help finding another player.

- a. Friend: I know you will lose your job if you lose the match this weekend - seriously, Andy can play rugby.

⁵Note that evidential adverbs in questions and epistemic adverbs in all contexts are not considered natural in sentence-initial position in English. However, this does not impact on the proposed analysis.

Interpretation: I am being serious/?take me seriously - both express that the speaker is being serious

- b. Manager: If we lose the match at the weekend I will lose my job - seriously, can Andy play rugby?

Interpretation: The speaker's question is serious/the answer from the addressee must be serious - addressee orientation remains despite context

Note that the shift is not related to the grammatical function of the arguments within the proposition:

- (16) a. It looks like it will rain tomorrow. (Speaker-orientation)
b. Does it look like it will rain tomorrow? (Addressee-orientation)

In (16), the evidential meaning of the matrix clause orients to the speaker in declaratives and the addressee in interrogatives. The phenomenon also seems to be robust cross-linguistically, as judgements in French and Greek mirror those in the English cases.

The data from embedded contexts is less clear, however. As noted by Banfield (1982) and Rooryck (2001), speech act markers such as parentheticals are not permitted in embedded contexts. Initially, it seems that illocutionary adverbs are permitted:

- (17) a. Dima told Nathan that Jessica seriously wants to come to the party.
b. Dima told Nathan that Jessica honestly spoke to Paul.

However, in these contexts the adverbs are interpreted as VP adverbs which modify the verb and its arguments rather than the sentence as a whole; *seriously* is interpreted as a degree adverb in (17a) and *honestly* as a manner adverb in (17b). This is supported by the French data, in which (18) is ungrammatical because *sérieusement* cannot be interpreted as a VP adverb:

- (18) ?*Guillaume dit à Jean que Marie voulait sérieusement aller à la fête de Martine
Guillaume says to Jean that Marie wanted seriously to-go to the party of Martine
'Guillaume says to Jean that Marie seriously wanted to go to Martine's party.'

Note that in the above examples, *seriously* and *honestly* are not sentence-initial. They can precede embedded clauses when pronounced with comma intonation and retain their illocutionary meaning, but this appears to degrade the sentence:

- (19) a. ?Dima told Nathan that, seriously, Jessica wants to come to the party.
b. ?Dima told Nathan that, honestly, Jessica spoke to Paul.

However, evidential adverbs like *allegedly* and the epistemic adverb *definitely* can occur in both matrix and embedded clauses:

- (20)
- a. Marie allegedly gave her lottery winnings to charity.
 - b. John told Harry that Marie allegedly gave her lottery winnings to charity.
 - c. Marie definitely will be at the party
 - d. John told Harry that Marie definitely will be at the party

It seems that the adverb searches for the appropriate discourse participant in the clause directly above its own. The overt performative structure of the matrix CP in (20d) contains the speaker and addressee to which embedded adverbs orient. This is confirmed by informants in English, French and Greek. Furthermore, the adverb in the matrix clause orients to the discourse participants. (20c) and (20d) are repeated below, with the adverb's orientation indicated using subscript:

- (21)
- a. Marie definitely_{speaker} will be at the party.
 - b. Will Marie definitely_{addressee} be at the party?
 - c. John told Harry that Marie definitely_{John} will be at the party.
 - d. John asked Harry whether Marie will definitely_{Harry} be at the party.
 - e. Amy told Ben that Charlie told David that Emma told Flora that Marie will definitely_{Emma} be at the party.
 - f. Amy asked Ben whether Charlie asked David whether Emma asked Flora whether Marie will definitely_{Flora} be at the party.

The examples show that the adverb orients to the most minimally distant 'speaker' in declarative cases and the most minimally distant 'addressee' in interrogative cases. It is not influenced by the grammatical status of the arguments in its clause relative to the discourse participants, and the embedded subject does not act as an intervener. These intuitions are confirmed by the felicity of the continuation shown below:

- (22)
- a. John told Harry that Marie definitely_{John} will be at the party, even though she's not sure she'll make it.
 - b. #John told Harry that Marie definitely_{speaker} will be at the party, even though she's not sure she'll make it.
 - c. #John told Harry that Marie definitely_{Marie} will be at the party, even though she's not sure she'll make it.

It will be proposed that, by analogy with the overt performative structure of the matrix clause, the discourse participants are represented as the arguments of a covert speech act head above matrix CP in section 4.2. In section 5 the question of why the matrix object is not a possible antecedent in (22a) despite being a valid speech act participant will be addressed.

4 A syntactic approach

I shall now present a structural representation of adverbs which shift, showing that their syntax is key to their interpretation. The syntactic subject of the adverb will be shown to be coreferential with a syntactic representation of the discourse participants above matrix CP. This will provide an elegant and minimal account for orientation shifting in adverbs, which falls out of independently motivated syntactic principles.

4.1 The adverb's argument

It is uncontroversial, following Bellert (1977), to suggest that adverbs which interface with the discourse can take semantic arguments. Bellert suggests that illocutionary adverbs, for example, take two arguments: the speaker and the content—specifically the proposition—of the sentence (1977, p.349). The adverbs in question here are outside of the proposition expressed in the sentence because they express the speaker's attitude to that proposition; specifically, these adverbs relate a property (seriousness, frankness etc.) to an entity with respect to the proposition expressed in the main clause.

The entity and the property are represented in the syntax and are pronounced overtly in combination as the adverb itself. In order to satisfy the Projection Principle (Chomsky 1981), which requires that the properties of lexical items must be preserved during the derivation of a sentence, the entity should also be represented itself, as a nominal element in the subject position of (in this case) the adverbial projection. However, the adverb cannot assign case to this subject, so the subject is never realised overtly. It is the only syntactic argument that the adverb has, because the adverb does not take the sentence as an argument, but the adverbial projection modifies it. These characteristics suggest that the adverb's subject is therefore a PRO and is subject to the principles of control.

It is now important to examine the characteristics of the adverb's PRO, first taking contexts in which sentential PRO receives an arbitrary interpretation. If the adverb is sentence-initial, its PRO does not receive an arbitrary interpretation.⁶ However, when evidential adverbs are sentence-internal, their orientation is uncertain between speaker- and arbitrary-orientation:

- (24) a. Frankly_{speaker}, PRO_{arb} to love another person is PRO_{arb} to see the face of God.

⁶The adverb *definitely* also resists an arbitrary interpretation, despite not being able to front to a sentence-initial position:

- (23) a. PRO_{arb} To definitely_{speaker/*arb} find a partner, use a dating service.
b. *Definitely, PRO_{arb} to find a partner, use a dating service.

For the moment I assume that this is related to the special property of *definitely* which allows it to shift in the first place (cf. Bellert 1977). The details of this will be left for future work.

- b. Apparently_{speaker}, PRO_{arb} to commit a crime is PRO_{arb} to face censure.
- c. PRO_{arb} To reportedly_{speaker/arb} have committed a crime is PRO_{arb} to face censure.

If we take an approach to PRO_{arb} along the lines of Epstein (1984), we assume that PRO_{arb} is a variable bound by an unselective universal operator in the IP domain. This universal operator is therefore lower in the structure than the representations of the discourse participants which will be presented in section 4.2. As such, this operator will never be a valid antecedent for the sentence-initial adverb's PRO, which is in a position above IP. However, when the evidential adverb is sentence-internal, as in (24c), its PRO can either orient to the discourse participants or to the universal operator. I propose that speaker-orientation is still (marginally) available, even though the universal operator is closer, because the adverb's PRO is sensitive to the discourse participants and as such will try to orient to a discourse participant in every case. As a result, the orientation of the adverb's PRO is unclear in this case.

Turning to contexts of 'non-obligatory' control (cf. Roberts (1997)), the adverb's PRO, like other typical PROs, is sensitive to such contexts. In clauses containing wh-movement, typical PRO does not require a controller:

- (25) John asked [how [PRO to shave oneself]] (Robert's (38), 1997, p.138)

In wh-questions, the adverb's PRO must take one of the discourse participants as a controller, but it is ambiguous between the SPEAKER and ADDRESSEE:

- (26) a. Seriously_{speaker/addressee}, who can play rugby?
 b. Sérieusement_{speaker/addressee}, qui est fort en foot?
 Seriously, who is strong at football?
 'Seriously_{speaker/addressee}, who is good at football?'

(French)

- c. Sovara_{speaker/addressee}, ti mporis na peksis;
 Seriously, what can.2sg SBJ play
 'Seriously_{speaker/addressee}, what can you play?'

(Greek)

It is clear then that the adverb's PRO subject is much like any other example of PRO, except that it is particularly sensitive to the discourse participants.

Returning to simple declarative and yes-no interrogatives, the data in section 3.2 suggest that PRO is controlled by a representation of the discourse participants (which one depends on the clause type). These are not standalone cases; the following sentences whose adjuncts contain PRO subjects which seem to be controlled by the discourse participants:

- (27) a. After $\text{PRO}_{\text{speaker}}$ pitching the tents, darkness fell quickly.
 b. After [my/our/?your/*her/*his] pitching the tents, darkness fell quickly.

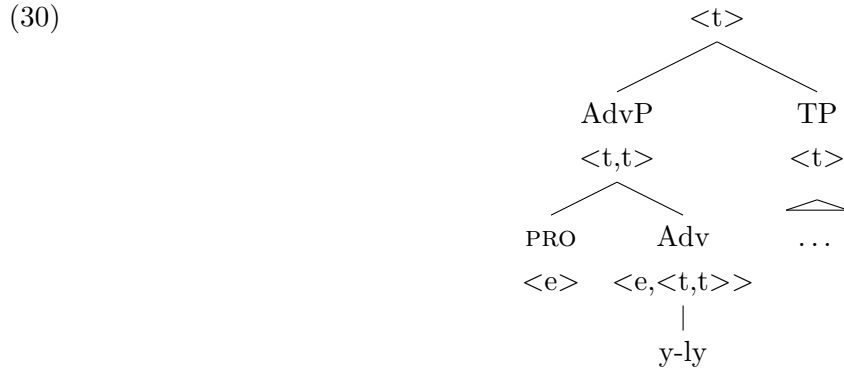
(from Boeckx & Hornstein (2004, pp.441-442))

- (28) a. John said that after PRO_{John} pitching the tents, darkness fell quickly.
 b. John_i said that after [*my/?our/?your/his_i/*his_j/their] pitching the tents, darkness fell quickly.

In terms of compositionality, the PRO merges with the adverb before the adverb merges with the larger structure. This is because the adverb does not actually modify the proposition, but expresses the discourse participant's attitude towards it. Semantically, the adverb is a function mapping individuals to propositional modifiers. The entire adverbial projection (i.e. the expression of the discourse participant's attitude) then modifies the sentence, because the proposition's meaning and truth values are the same regardless of the discourse participant's attitude. This is illustrated in (29):

- (29) If p is a proposition, x is an individual, and **y-ly** is an argument-taking adverb, then:
 a. **y-ly**-for(x), (p)
 b. #For (x), **y-ly**(p)

An analysis of the semantic types of the elements involved suggests that PRO merges with the adverb first:



I have already described about how the adverb's PRO argument must be controlled by an antecedent which is a discourse participant higher in the structure than PRO itself;⁷ therefore, the discourse

⁷I assume that this is not a case of backward control, as the matrix subject does not control the PRO in matrix cases, even if it is a discourse participant:

- (31) a. You are definitely_{speaker} the best footballer in the class.
 b. #You are definitely_{you} the best footballer in the class.

participants too must be represented above matrix CP.

This is similar to Ross (1970) and Speas and Tenny's (2003) proposals and will elegantly explain this shifting phenomenon. The key difference between Ross's proposal and mine is based around what constitutes the head of the covert structure. Ross claimed a direct correspondence between affirmative declarative sentences both with and without performative constructions, and that the latter were formed by the deletion of the performative construction from the former, as shown below:

(32) I'll be there = (e.g.) ~~I promise you that~~ I'll be there

This was rejected by Fraser (1974), Gazdar (1979), Leech (1983) *inter alia* for a number of well-argued reasons, for example the fact that not all performative verbs have first-person subjects and second-person objects, that there is no 1:1 mapping between utterances and types of speech act, and that it is possible to stack performatives, posing further problems for the interpretation of the (most) subordinate clause.

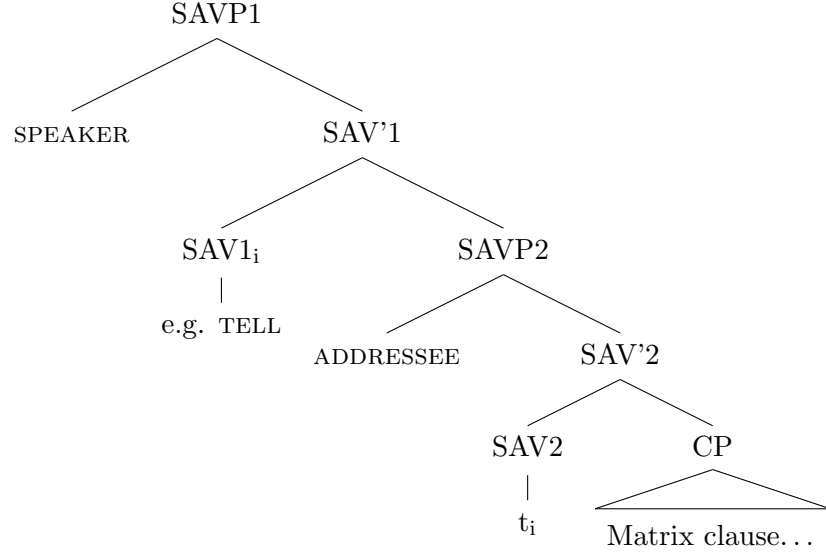
I therefore wish to be clear that I am not proposing that every speech act is a half-pronounced performative structure. Instead, I will propose that there is a covert operator which relates to the matrix clause type and which selects its arguments on this basis. I will be representing this operator using English verbs of communication, but this should be interpreted solely as a representation. I will elaborate on this idea in section 5.

4.2 The covert Speech Act structure

To account for the facts in section 3.2, I propose a covert speech act verb (SAV) which subcategorises for a certain configuration of arguments depending on the clause type it precedes. The SAV is the head of a covert Speech Act Verb Phrase (SAVP) above the matrix CP.

The structure, as a first analysis, is that of a three-place predicate, with the SPEAKER in the specifier (subject position) of the whole projection, the sentence/matrix CP as the complement of the SAV (direct object position) and the ADDRESSEE as the complement of the higher SAV (in the indirect object position):

(33)



The advantage of this structure is that it is an accurate analogy for the relationship between the discourse participants and the discourse itself based on verbs of communication. Furthermore, it does not trigger embedded phenomena in the matrix CP because it has the same relationship with the matrix CP that a parenthetical would have with the matrix CP in the case of direct quotation. As such, the SAVP represented as [Speaker say] in (34a) is roughly analogous⁸ to “Louise said” in (34b):

- (34) a. [Speaker say] It is cold outside today.
 b. Louise said, “It is cold outside today.”

However, it does not yet fully reflect the difference between different clause types; specifically the difference between declarative and interrogative clauses.

4.2.1 Differences in declarative and interrogative SAV structures

I will first refine the structure for declarative SAVPs. It is both intuitive and empirically sound to suggest that only speakers and not addressees are obligatorily structurally represented in declarative SAVPs. Intuitively, declarative sentences may be uttered without an addressee, as they can be stated to no-one in particular. They may also be addressed to an audience which is not one definable entity or set of entities, or to those may not even be the intended audience for the sentence. This is not the case with interrogatives; although the speaker and the addressee may in some contexts be the same person, there is always a specific addressee in mind, otherwise the act

⁸The extent to which this analogy holds is not yet clear, especially given recent work in child developmental literature which suggests that children respond differently in Theory of Mind tasks when the direct speech of the characters in the task is framed with a parenthetical and when it is not (De Villiers 2014). I will not therefore claim that the analogy is precise, but will leave the details for further research at this point.

of asking a question might not provide the speaker with the new information s/he requires.

Deleted arguments provide empirical support to these intuitions; consider the following sentences:

- (35) a. The boss wants to meet ~~me~~/~~?you~~/~~us~~ at 3pm.
 b. Does the boss want to meet ~~me~~/~~you~~/~~us~~ at 3pm?
- (36) (Context: a company only accepts payments in pounds sterling)
 a. The client tried to pay ~~me~~/~~*you~~/~~us~~ in dollars!
 b. Did the client try to pay ~~me~~/~~you~~/~~us~~ in dollars?

In (35a) and (36a), the preferred reading is that the deleted argument is co-referential with a first-person argument. The second-person reading is strange and dispreferred, if not outright rejected by informants (cf. also Ross's examples in (9)). Informants report that further context is required to readily construe a second-person reading. In contrast, both first- and second-person readings are available in (35b) and (36b).

I will therefore modify the structure in (33) to show that only **SPEAKERS** are represented in declarative SAVPs:⁹

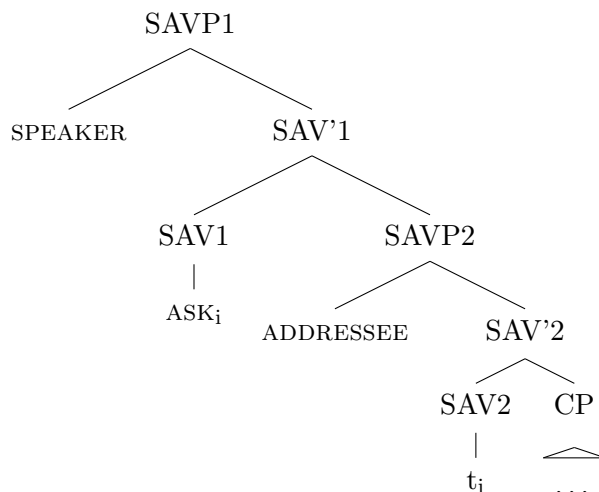


The SAV in declarative SAVPs is something like **ASSERT**. This reflects the fact that, unlike **TELL**, there are only two arguments in the SAV structure of declarative sentences - there is no **ADDRESSEE** as an indirect object argument. This means that, as the only available argument, the **SPEAKER** controls the **PRO** argument of the shifting adverb in **CP**.

However, in interrogative SAVPs both the speaker and addressee must be present, as example (35) shows. This means that the SAV in interrogative SAVPs is a three-place predicate as in (33), but the SAV is something like **ASK** to show that this structure solely applies in interrogative contexts:

⁹It could be argued that addressees are also represented in declaratives, as example (36a) is acceptable with the first-person plural pronoun 'we'. However, in this case the speaker is speaking for other people and including others in his perspective. As such, the other participants included within 'we' are not **ADDRESSEES**, and may not even be present for the discourse at all, but are merely people who share the same relevant co-ordinates as the **SPEAKER**. Furthermore, languages which show a distinction between the exclusive and inclusive first-person plural pronoun illustrate the optionality of including the addressee.

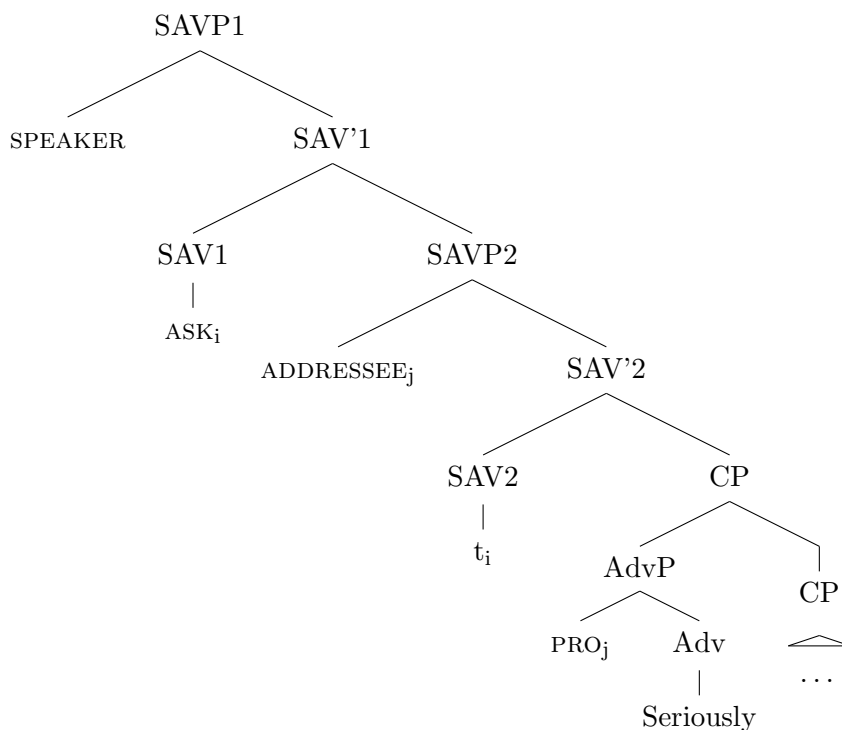
(38)



This now means that the adverb's PRO has two potential antecedents to choose from. However, the ADDRESSEE when present is hierarchically closer than the SPEAKER. Therefore, following Rosenbaum's (1970) Minimal Distance Principle (MDP), the ADDRESSEE always controls the adverb's PRO in interrogative clauses.

Finally, to illustrate how the adverb fits into the picture, it is part of the sentence so will be inside the matrix CP:

(39)



It is not particularly important for this account where the adverbs are base-generated. According

to Cinque’s (1999) hierarchy, the adverbs in question would be generated in the following order:¹⁰

- (41) Speech act (illocutionary) >Evaluative >Evidential >Epistemic

Therefore, given that illocutionary and evidential adverbs shift, whereas (most) epistemic adverbs do not, whether an adverb shifts is not related to how high in this hierarchy it is generated.

4.2.2 Embedded contexts

However, as noted in section 3.2, shifting adverbs generated lower in the hierarchy shift in embedded as well as in matrix contexts (cf. example (21)). This is also true in both French and Greek. In contrast, illocutionary adverbs can only shift if pronounced with comma intonation and then the sentence is still considered degraded. The paradigm is illustrated in the examples below:

- (42) Illocutionary adverbs
- a. *George told Nikki that honestly_{George} Norman had won the race (only VP-adverb reading available)
 - b. ?George told Nikki that, honestly_{George}, Norman had won the race
 - c. *George asked Nikki whether honestly_{Nikki} Norman had won the race
 - d. ?George asked Nikki whether, honestly_{Nikki}, Norman had won the race
- (43) Evidential adverbs
- a. George told Nikki that Norman had allegedly_{George} won the race
 - b. George asked Nikki whether Norman had allegedly_{Nikki} won the race
- (44) ‘Definitely/perhaps’
- a. George told Nikki that Norman had definitely_{George} won the race
 - b. George asked Nikki whether Norman had definitely_{Nikki} won the race
 - c. George told Nikki that Norman had perhaps_{George} won the race
 - d. George asked Nikki whether Norman had perhaps_{Nikki} won the race

But why should comma intonation improve (if not render grammatical) the examples in (42b) and (42d)? Emonds (1976, p.61) suggests that comma intonation around an illocutionary adverb in an embedded clause allows the illocutionary adverb to be distributed as a parenthetical throughout

¹⁰I have not looked at evaluative adverbs in this paper as their behaviour is markedly different; they are not available in questions and are much more susceptible to shifting with the context even in declarative sentences, for example:

- (40) a. There will be a prize given for the winner of the race. Fortunately [for me], I will win.
b. There will be a forfeit given to the winner of the race. Fortunately [for everyone else], I will win.

I will leave these facts for future research.

that embedded clause. Furthermore, Emonds draws a parallel between parentheticals pronounced with comma intonation and direct speech contexts (pp.23-25, p.44). Specifically, he notes that comma intonation is used to convey direct speech, creating a partition between the parenthetical and the direct quotation, across which the third-person matrix subject of the parenthetical is referred to using the first-person in the quotation:

(45) George said to Nikki, “Honestly_{George}, Norman won the race. I_{George} saw it happen.”

With the caveat that cases such as (42b) and (42d) are only marginally acceptable, it seems that by pronouncing the adverb with comma intonation, the indirect quotation is given a ‘direct-quotation-like’ reading. This creates a kind of embedded root environment, thereby marginally permitting illocutionary adverbs (clearly a root phenomenon) despite their presence in an embedded clause. Quasi-quotational contexts, to use the terminology of Bayer (2004), may also be induced in other syntactic environments. In Hiberno-English (McCloskey 2006), some North West English varieties and African-American English (Jill De Villiers, p.c.) subject-auxiliary inversion, another root phenomenon, is possible in complement-less embedded interrogative clauses (note the contrast of acceptability of (46c)):¹¹

- (46) (from McCloskey (2002), in Bayer (2004))
- a. He asked me would I cook tea.
 - b. I asked Jack was she in his class.
 - c. *I asked Jack whether was she in his class.

As expected, speakers who accept (46) also accept illocutionary adverbs with shifted interpretations in such contexts:

- (47)
- a. He asked me, seriously_{me}, would I cook tea.
 - b. I asked Jack, honestly_{Jack}, was she in his class.

Note that comma intonation is obligatory (just like in matrix clauses) but these examples are not degraded for speakers of these dialects, unlike the examples in (42b) and (42d).

In sum, illocutionary adverbs are root phenomena that are acceptable in embedded root environments which promote a quasi-quotational context. Such contexts can be created through the use of comma intonation and, in dialects such as Hiberno-English, complement-less embedded interrogatives. However, the MDP still holds, as the real-world speaker and addressee are maximally distant and therefore not available to control the adverb’s PRO in these cases.

¹¹Further evidence may be available from embedded V2 contexts in languages such as German (Bayer 2004) and Afrikaans (Biberauer 2002).

5 The role of logophoricity

I have shown that the adverb's argument is a PRO, and therefore it must be controlled. But what kind of control is the adverb's PRO subject to? Williams (1992) proposes two forms of control for understood subjects in adjuncts. Firstly, Predicative Control (PC) involves an adjunct which is directly predicated of a subject and strict locality requirements are imposed. He judges that there is no PRO in these cases (contra Chomsky (1981) and Stowell (1983)) as the adjunct directly assigns its theta-role to the subject, for example in the following sentence:

(48) John_i arrived asleep_i

However, Logophoric Control (LC) has no locality requirement and theta-role assignment is separate from indexing. It is therefore considered to involve a PRO and there is a requirement in these cases for the antecedent to be a discourse participant.

It appears that LC is a more appropriate analysis than PC for the kind of control seen in illocutionary and evidential adverbs. An argument against PC is that there is no full DP subject in the same clause as the adverb to which it can assign its theta role (i.e. instead of assigning it to a PRO subject). Even when the matrix subject precedes the adverb (presumably due to some kind of movement) it is clear that the adverb is not predicated of this subject by the interpretation that it receives (cf. (22)). This is because the matrix subject has already been assigned a theta-role by the matrix verb and because the matrix subject is not necessarily a discourse participant.

Furthermore, the cases of PC described by Williams which involve adverbs all involve VP adverbs. It is crucial to note that illocutionary and evidential adverbs are *not* simply a kind of manner adverb. This is illustrated by the following paraphrases, of which (49a) does not paraphrase the meaning of the illocutionary adverb whereas (49b) and (49c) do:

- (49) “Seriously, you can play rugby.”
- a. #I said seriously [that you can play rugby]
 - b. I said thus: “Seriously, you can play rugby.”
 - c. I convey that I am serious about the proposition that you can play rugby.

Importantly, the adverb and its meaning are clearly part of the utterance rather than part of the utterance context itself (though other linguistic and non-linguistic factors may make it seem to be so). It has already been shown how the adverb orients to the discourse participants which are represented in a SAV structure above the matrix CP. It is clear therefore that the adverb's subject cannot be and is not locally controlled (as PC would require) and that it is sensitive to the discourse participants (as required by LC). Now it remains to be determined how the adverb's PRO finds its antecedent from among the discourse participants provided by the SAV structure. This section details how the PRO in shifting adverbs is sensitive to the logophoric centre of the speech act.

5.1 The logophoric centre

First, the term ‘logophoric centre’ must be defined. Sells (1987, p.457) proposes that the logophoric ‘self’ is the individual “whose mental state or attitude the content of the proposition describes”; so in the terms already used in this paper, the individual to whom the adverb orients. However, I will avoid the term ‘self’ as I will make some assumptions which may not be originally intended by Sells, and will refer to the logophoric centre instead.

It is necessary to define the logophoric centre because the data in section 3.2 show that there are arguments which precede the adverb but cannot serve as an antecedent for the PRO. For example, the subject of the clause in which the adverb resides may precede the adverb, but this subject does not act as an intervener between the adverb and the discourse participants:

- (50) a. Seriously_{speaker}, you can play rugby.
b. You seriously_{speaker/*you} can play rugby.

Moreover, neither the embedded subject nor the matrix object block control of PRO by the matrix subject in embedded declaratives, as illustrated in (22a) and (22c). Two consequences follow; firstly, the adverb’s PRO is not simply controlled by the nearest antecedent, but by the nearest *viable* antecedent. Secondly, the licensing factor which determines viability is common to both the discourse participants and the matrix arguments of a verb of communication.

I propose that logophoricity is this extra factor because adverbs in the same clause as an intent verb orient to the intent verb’s subject.¹² This is because verbs such as ‘intend’ express the mind and consciousness of their subject, so the subject becomes the most minimal viable antecedent:

- (52) John told Harry that Marie definitely_{Marie} intends to go to the party, even though she’s not sure she’ll make it.

Note that logophoricity is not a perpetually available mechanism; it exists because there is a speech act or an act of expression of some kind being performed. In this way, it does not solve the problem of adverb orientation alone, but falls out of the structures that have been proposed in the previous sections. Furthermore, it must be highlighted that, in the case of reported speech, there will be two logophoric centres; one for the reported speech act and one for the main speech act. This fact is important because it means that the adverb is limited by the speech act which it modifies; an

¹²Note that this is only the case when the adverb follows the intent verb; therefore *seriously* never orients to the subject of a intent verb:

- (51) a. Seriously_{speaker}, Marie intends to go to the party.
b. Frankly_{speaker}, Marie intends to go to the party.
c. Marie seriously intends to go to the party.
d. ?Marie frankly intends to go to the party.

In (51c), *seriously* can only be interpreted as a degree adverb.

embedded adverb may not take wide scope over the entire sentence.

I shall now show that the SAV operator determines which of its arguments is the logophoric centre of the main speech act in a similar way to overt lexical verbs of communication, and how this interacts with a theory of questions.

5.2 The SAV as a logophoric verb

Verbs of communication have lexical properties which determine the kind of speech act they introduce. For example, *say* and *tell* subcategorise for declarative CP complements, whereas *ask* and *inquire* take interrogative ones. They also differ as to whether they subcategorise for an indirect object (an addressee).

As noted above, in sentences with a matrix verb of communication have two logophoric centres: one for the main speech act and one for the embedded speech act. Sells notes that it is a lexical property of a verb of communication that its logophoric centre is predicated of its subject (1987, p.457). If a matrix verb of communication determines the logophoric centre for the embedded speech act, then by analogy the SAV will determine the logophoric centre for the main speech act. This proposal again works on the principle that it is the syntax and semantics of speech acts which determine adverb orientation, specifically the syntactic presence of the discourse participants above the matrix CP, and that the role of logophoricity here falls out of this fact.

Sells (1987, pp.452, 454) refers to verbs of communication such as *say* and *ask* as logophoric predicates. The addition of a logophoric predicate above a non-logophoric statement induces logophoric behaviour in the (newly) subordinate clause, as seen in Icelandic (examples from Sells 1987, p.452):

- (53) a. *Ólafur hefur ekki enn fundith vinnu, sem sér_{Olaf} likar
 Olaf has not yet found a-job that self_{Olaf} likes
 ‘Olaf has not yet found a job that he_{Olaf} likes.’
 b. Jón segir ath Ólafur hefur ekki enn fundith vinnu, sem sér likar
 Jon said that Olaf has not yet found a-job that self_{Jon/Olaf} likes
 ‘Jon said that Olaf has not yet found a job that he_{Jon/Olaf} likes.’

The logophoric pronoun *sér* cannot be bound by Olaf in (53a). However the introduction of the logophoric predicate “Jón says” permits either Jón or Olaf as the antecedent of *sér* in (53b). This fact, with others as presented in Sells’s work, seems to show that the CP containing logophoric elements must be subcategorised for by a logophoric verb in order for the logophoric element to be correctly bound (Sells 1987, p.473).

I therefore propose that for a shifting adverb to be licensed, it must be contained within a CP selected by a logophoric element. What is more, this logophoric element determines the logophoric centre of the speech act.

In terms of how the logophoric element (the SAV operator) determines the logophoric centre,

Kuno (1987) suggests that logophoric verbs mark both their speaker and addressee arguments as [+log], whether they are covert or overt. As the proposal in this paper requires that there is only one logophoric centre and therefore one logophorically active argument in each speech act (cf. Speas and Tenny 2003), this requires examination.

Kuno claims that a full non-pronominal DP in the embedded clause cannot be co-indexed with either of the arguments of a logophoric verb (examples and judgements from Kuno 1987, p.110):

- (54) a. *That John was crazy was just one of the things Mary said to him_{John} (John is the addressee of Mary's remarks)
 b. That John was crazy was just one of the things Mary said about him_{John} (John is not the addressee of Mary's remarks)

However, it may not truly be logophoricity that is at play here. From a theoretical standpoint, Sells (1987) is very clear that there is no such primitive role as 'speaker', but that the speaker of a sentence takes on one or more of the logophoric roles he proposes, namely 'source', 'self' or 'pivot'. This is because crosslinguistic phenomena such as the orientation of logophoric pronouns in embedded indirect speech cannot be explained using an "unanalysed notion of speaker", but can be accounted for by a more primitive and ultimately flexible notion of 'source' (Sells 1987, p.462). This does not mean that the speaker cannot be in some way represented in the syntax, but that it is not inherently logophoric; ditto the addressee.

Consider what it would mean for the addressee to have a logophoric role in (54). The addressee is not responsible in terms of form, content or point of view for any part of these sentences, and does not contribute to the setting of indexicals. It would be predicted that the addressee would be an anchor for some kind of meaning or coreference if it were in some way logophoric. However, the meaning and interpretation of sentence (54b) does not change whether Mary addresses it to Fred, George, Jane, or any other addressee. The meanings of the speech acts contained within (54a) and (54b) are the same. A small-scale survey of English native speakers also confirmed no differences in grammaticality between (54a) and (54b).

Furthermore, Kuno himself notes distinctions between the speaker and the addressee, for example in 'as for X-self' cases (1987, p.129):

- (55) a. As for myself, I won't be invited
 b. ??As for yourself, you won't be invited
 c. *As for herself, she won't be invited.

Kuno argues that this is because the speaker's logophoric role is more acceptable than the addressee's logophoric role as an antecedent for reflexives. However, this seems stipulative, because both logophoric roles were deemed to motivate the same levels of ungrammaticality in (54), and because Kuno does not explain why and in what ways they are different. It is more motivated

to suggest that the addressee is marginally available to some speakers in ‘as for X-self’ contexts because of the high degree of saliency of the addressee in the context compared with a third person, but that its marginal status and its variability across idiolects suggests that it is not a syntactic antecedent in the same way that the speaker is.

It has therefore been established that logophoric verbs of communication which take declarative CP complements assign one logophoric role to their external argument (their speaker). Furthermore, only SPEAKERS and not ADDRESSEES are subcategorised for by the declarative logophoric SAV. This is because addressee co-ordinates are not taken into account and are not available for binding in the matrix CP (at least in English), so their inclusion would be trivial. To represent the declarative SAV head in the case of English, the transitive declarative verb *assert* is used because it takes two arguments (the speaker as its external argument and the sentence as its internal argument). *Assert* could equally well be labelled *say*, but not *tell* which subcategorises for three arguments and therefore is not a possible representation of the declarative SAV. As for the interrogative SAV head, *ask* is ditransitive and takes three arguments (the speaker, the sentence as its direct object and the addressee as its indirect object).

However, only through a closer examination of the nature of questions are two vital facts motivated: one, that both the speaker’s and addressee’s co-ordinates are represented in questions; and two, that it is a lexical property of interrogative verbs of communication that the logophoric centre is predicated of its indirect object, i.e. the addressee.

5.3 The logophoric centre in questions

It is intuitive to suggest that the speaker is the logophoric centre of a simple declarative sentence; it is expected that the speaker can only share the contents of his or her own mind.

But what about questions? Firstly, when a speaker utters an interrogative sentence, this sentence does not straightforwardly constitute a proposition. It represents a certain piece of information which the speaker requires but does not know and cannot therefore express the speaker’s mind. Instead, the logophoric centre of the sentence becomes the person who will provide an answer, namely the addressee.

A common conceptualisation of questions is based on the fact that knowing the meaning of a question means knowing the meaning of the answer. From Hamblin (1973) and Karttunen (1977) to Groenendijk & Stokhof (1994), questions have been conceptualised as a set of possible or true answers, from which the addressee must pick the one which corresponds to his/her situation. This must mean that in order to present an answer-set from which one answer may be chosen, the speaker does not evaluate not his/her own situation, which can offer the alternatives but not choose the true one. Instead, to use Groenendijk and Stokhof’s (1994) terms, the speaker chooses an addressee who s/he *considers* to have an epistemic situation which overlaps with the partition of the logical space that the speaker has created by asking the question.

Groenendijk & Stokhof (1994, p.1095) define answerhood according to a model M as follows (ϕ represents an answer and $? \psi$ a question):

$$(56) \quad \phi \models_M ? \psi \text{ iff } \exists w \in M : [\phi]_M \subseteq [?\psi]_{M,w}.$$

This means that in the speaker’s world, a set of alternatives can be provided which includes the answer, but this is as far as the speaker can go. The speaker therefore chooses an addressee whose world - as far as the speaker is aware - contains both the set of alternatives $[?\psi]_M$ and the information to select the true one.

It is crucial that, if the speaker wants to acquire a valid answer, then s/he is responsible for choosing an addressee who has the requisite epistemic situation. This actually means that the ADDRESSEE is identified and *constructed* by the speaker from the speaker’s best knowledge. Furthermore, shifts in indexicality in echo questions suggest that the speaker relinquishes the logophoric centre to the addressee in interrogatives. Banfield (1978, p.436) notes that echo questions typically reproduce the questioned speech verbatim as in (57) below:

- (57) A. George Eliot wrote Middlemarch.
 B. George Eliot wrote Middlemarch?
 B’.#Mary Ann Evans wrote Middlemarch?

However, if the original speech contains first- or second-person pronouns, these shift in echo questions:

- (58) Q. Would you prefer a cup of tea?
 A. Would I prefer a cup of téa?
 A’. *Would you prefer a cup of téa?

As Banfield suggests, this shows that the speaker is, in her words, “divorced” from the logophoric centre when asking questions (Banfield 1978, p.437).

Because the speaker constructs the ADDRESSEE from his/her own knowledge, s/he can misrepresent the ADDRESSEE’s co-ordinates without causing syntactic ungrammaticality. For example, in the discourse in (59), *definitely* orients to the ADDRESSEE B, even though the continuation of the discourse shows that the addressee herself may not be ‘definite’ about the proposition:

- (59) A. Did Margarita definitely_B go to your party?
 B. Presumably_B she came, because I saw her coat in the hall (but I didn’t see Margarita directly).¹³

¹³Littell et al. (2010, p.94) note that evidentials in Salish languages seem to weaken the interrogative force of the question containing them. This seems to be true of the adverbs discussed here too; (59) is primarily concerned with whether or not the speaker is definite, rather than whether Margarita went to the party. This suggests that the

The same case holds with Tibetan evidentiality markers. If the speaker presumes indirect evidence on the part of the addressee, but the addressee has direct evidence, the speaker's sentence is not ungrammatical, because it is based on the speaker's representation of the addressee's situation. However, the addressee's reply will be ungrammatical¹⁴ if s/he doesn't correct the evidential (example from De Villiers et al. (2009, p.35)):

(60) (context: Speaker A rings speaker B at home to ask if Tashi is there. Speaker A assumes that B will only have indirect evidence of this fact, but B is sat opposite Tashi during the call)

A. Bkra-shis nang-la yod-sa-red_B pas?

Tashi in EVID.IND Q
'Is Tashi in (can you tell)?'

B. *Nang-la yod-sa-red_B

In EVID.IND
'She is in (I can infer)'

B'. Nang-la dug_B

In EVID.DIR
'She is in (I can see her)'

Note that no (other) co-ordinates shift, and the addressee's co-ordinates do not in any way subsume those of the speaker. This is evident because the orientation of indexical pronouns is the same in questions as in statements. It is accounted for because the shift is not related to personhood or indexicality, but to how the type of SAV used subcategorises for the logophoric centre.

However, logophoric pronouns conform to predictions as they do shift orientation in questions. The Chinese logophoric pronoun *ziji*, when not bound in the sentence, is thought to be bound by the speaker (example from Chen (2009, p.466)):

(61) Zhe-ge mimi zhiyou *ziji*_{speaker} zhidao
This-CL secret only self know
'Only myself knows this secret'

However, in questions, the 'default' interpretation of *ziji* shifts to the addressee (example from Pan (2001)):

(62) *Ziji*_{addressee} wei-she-me bu qu ne?
self why no go Q
'Why doesn't yourself go?'

speaker predicts a certain response from the particular addressee on the level of the proposition contained in the answer, further suggesting that the speaker is interested in the addressee's understanding when asking the question.

¹⁴De Villiers et al (2009, p.35) report that if a speaker uses the wrong evidential for the evidence which s/he has, this is judged as a grammatical violation, not just pragmatic infelicity.

These facts show that the logophoric centre in questions shifts because the addressee’s situation, as created by the speaker, is the one at issue.

6 Conclusion

This paper shows that orientation shifting in high adverbs is a systematic and consistent phenomenon not only in English, French and Greek. The data show that high adverbs orient to a syntactic representation of the SPEAKER in declarative clauses, and that illocutionary and evidential adverbs, as well as *definitely* and *perhaps*, shift in yes-no interrogatives to orient to the ADDRESSEE. The phenomenon also occurs in clauses embedded under verbs of communication in the cases of evidential adverbs and *definitely* and *perhaps*. These adverbs orient to the matrix subject (‘speaker’ of the embedded clause) in declarative embedded clauses and to the matrix object (‘addressee’) in interrogative embedded clauses. Illocutionary adverbs, in contrast, are root phenomena which can only shift in matrix clauses or in embedded root contexts.

I have postulated two elements of structure to account for this phenomenon, both of which fall out of independently motivated principles. Firstly, illocutionary and evidential adverbs have PRO subjects which are sensitive to the logophoric centre of the clause. Following the MDP, this PRO is controlled by the most minimally distant logophoric centre which antecedes it, thereby determining the orientation of the adverb phrase. Secondly, the discourse participants are syntactically represented as the arguments of a covert speech act phrase above matrix CP. The SAV head determines which discourse participants are obligatorily present and which one is the logophoric centre of the speech act. Above declarative clauses, the SAV ASSERT takes two arguments; the SPEAKER, which the logophoric centre is predicated of, and the sentence uttered. Above interrogative clauses, the SAV ASK takes three arguments; the SPEAKER, the sentence uttered and the ADDRESSEE, which takes the role of logophoric centre in questions because the addressee is assumed by the speaker to have the requisite situation to provide an answer. There is one logophoric centre per speech act. In this way, in declaratives the adverb’s PRO is controlled by the subject of the verb of communication which subcategorises for the clause in which the adverb resides; in short, the speaker of the speech act of which the adverb is part. In interrogatives, the adverb’s PRO is controlled by the indirect object of the verb of communication which subcategorises for the clause in which the adverb resides; that is the addressee of the speech act of which the adverb is part.

This solution can therefore be summed up as follows: orientation is a case of predication. The adverb is predicated of a subject which is bound by a representation of a discourse participant. This solution is solid and elegant because it falls out of independently motivated principles such as the MDP and only postulates structure which has overt correlates across languages. It does not postulate features that are not empirically motivated, but instead focuses on structures and functional projections which have seen strong empirical support. Finally, it can offer with further development a solution to the persistent problem in Minimalist syntax of spelling out the highest

projections in a sentence without resorting to stipulative means.

Future directions for this research are several: the proposal must be extended to other clause types such as imperative clauses as well as to languages with grammaticalised means of expressing the moods and modalities of the high adverbs. The question of why certain adverbs are not available in questions, and how these adverbs constrain the answer-set available in the partition formed by questions must also be examined. Finally, the consequences of the SAV analysis for phase theory will be detailed.

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